Module 1 – SE - Overview of IT Industry

1.Research different types of internet connections (e.g., broadband, fiber, satellite) and list their pros and cons.

Fiber Optic

- Pros: Fastest speeds, most reliable, low lag.
- Cons: Limited availability, can be expensive.

Cable

- Pros: Very fast, widely available.
- Cons: Slows down during peak hours (shared bandwidth).

DSL

- Pros: Uses existing phone lines, cheaper.
- Cons: Slower speeds, quality drops with distance.

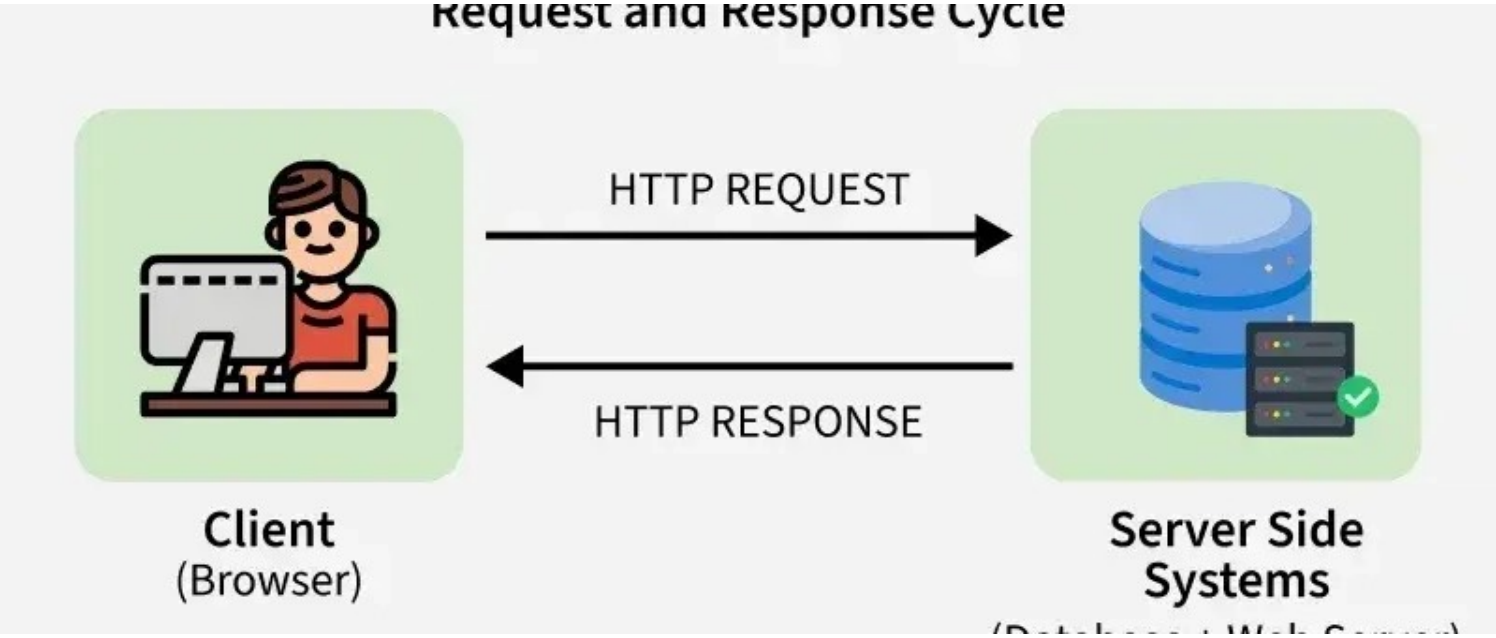
Satellite

- Pros: Available in remote/rural areas.
- Cons: High latency (lag), affected by weather, data caps.

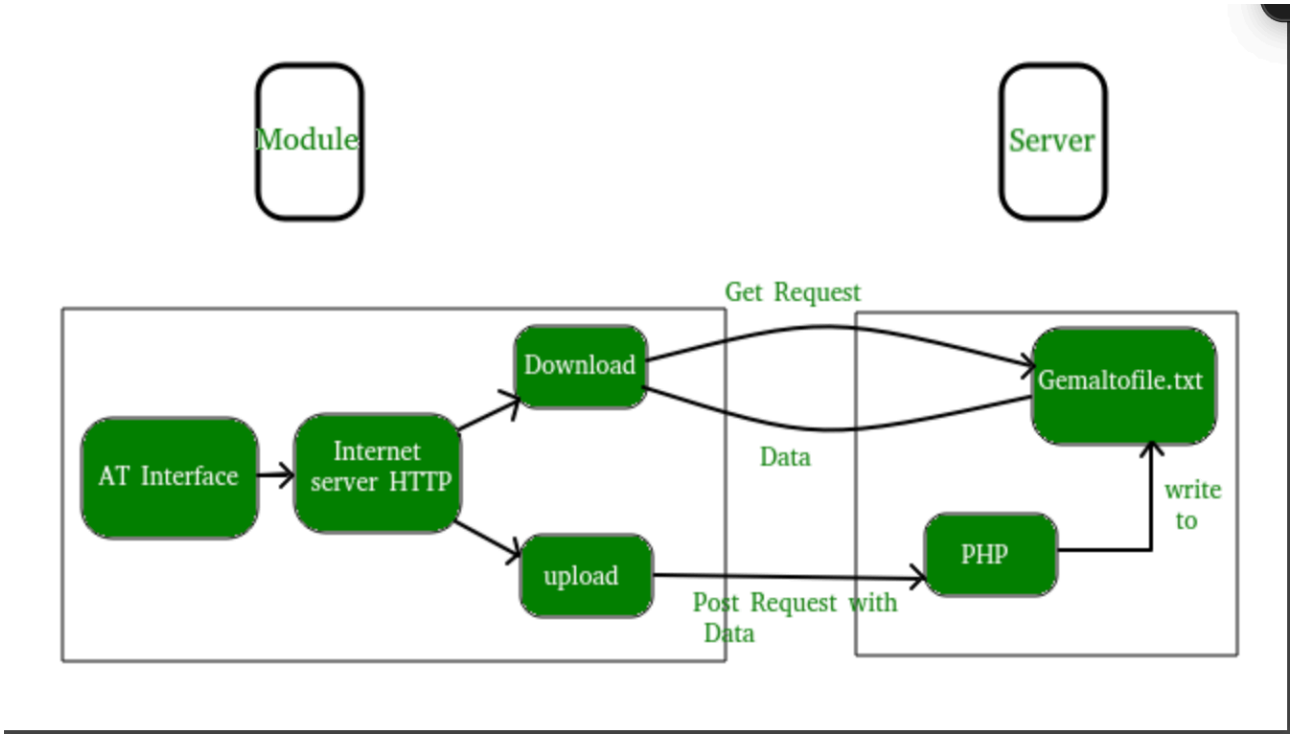
Fixed Wireless / 5G

- Pros: Easy setup, no wires needed.
- Cons: Speed varies by signal strength and obstacles.

2.Research and create a diagram of how data is transmitted from a client to a server over the internet



3.: Design a simple HTTP client-server communication in any language



4.Identify and explain three common application security vulnerabilities. Suggest possible solutions.

A.Three common application security vulnerabilities are injection attacks (like SQL injection), broken authentication, and security misconfigurations.

- **<u>Injection attacks solution</u>** : -Use prepared statements and parameterized queries. -Perform strong input validation to reject any data that doesn't meet expected formats. -Enforce output encoding to properly format data before it is displayed in a user interface

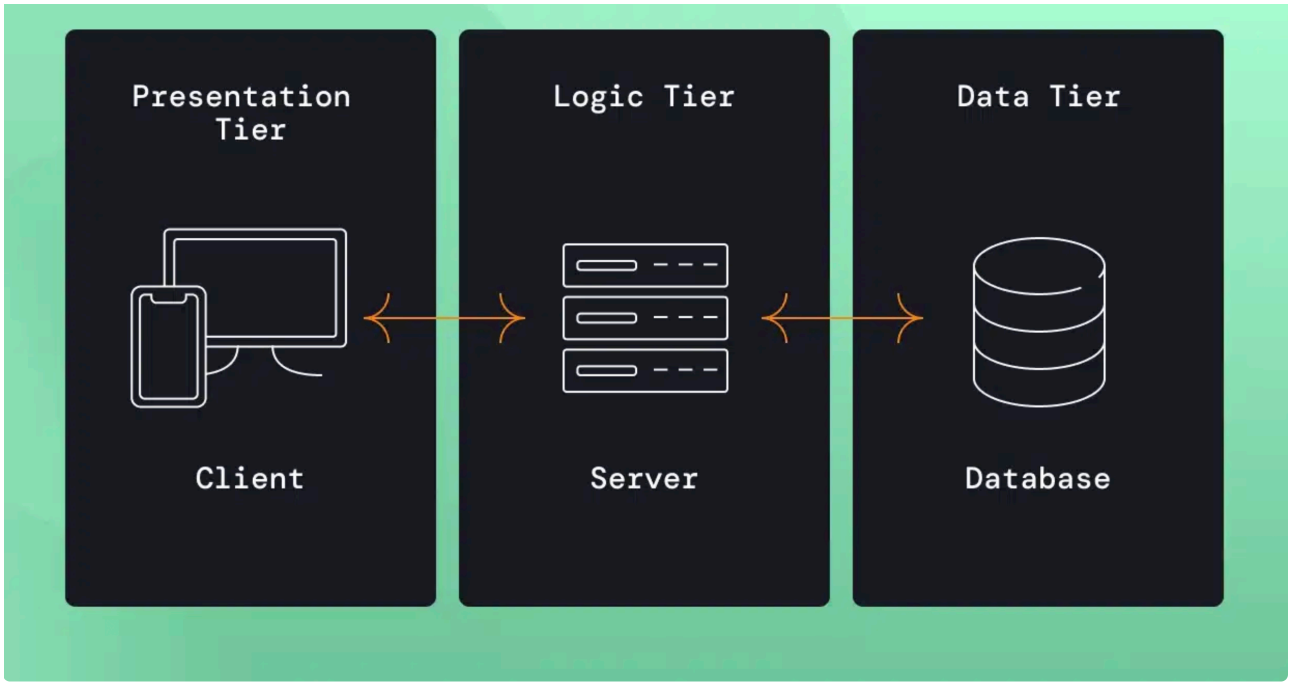
- **<u>broken authentication solution</u>** : -Enforce multi-factor authentication. -Implement a strong password policy that includes complexity requirements and periodic changes. -Ensure secure session management with secure cookies and by invalidating sessions after a period of inactivity.

- **<u>security misconfigurations :</u>** -Remove unnecessary features, services, and accounts, especially default ones. -Regularly patch and update all components, including the operating system, framework, and libraries. -Ensure security settings are properly configured and security headers are set to secure values.

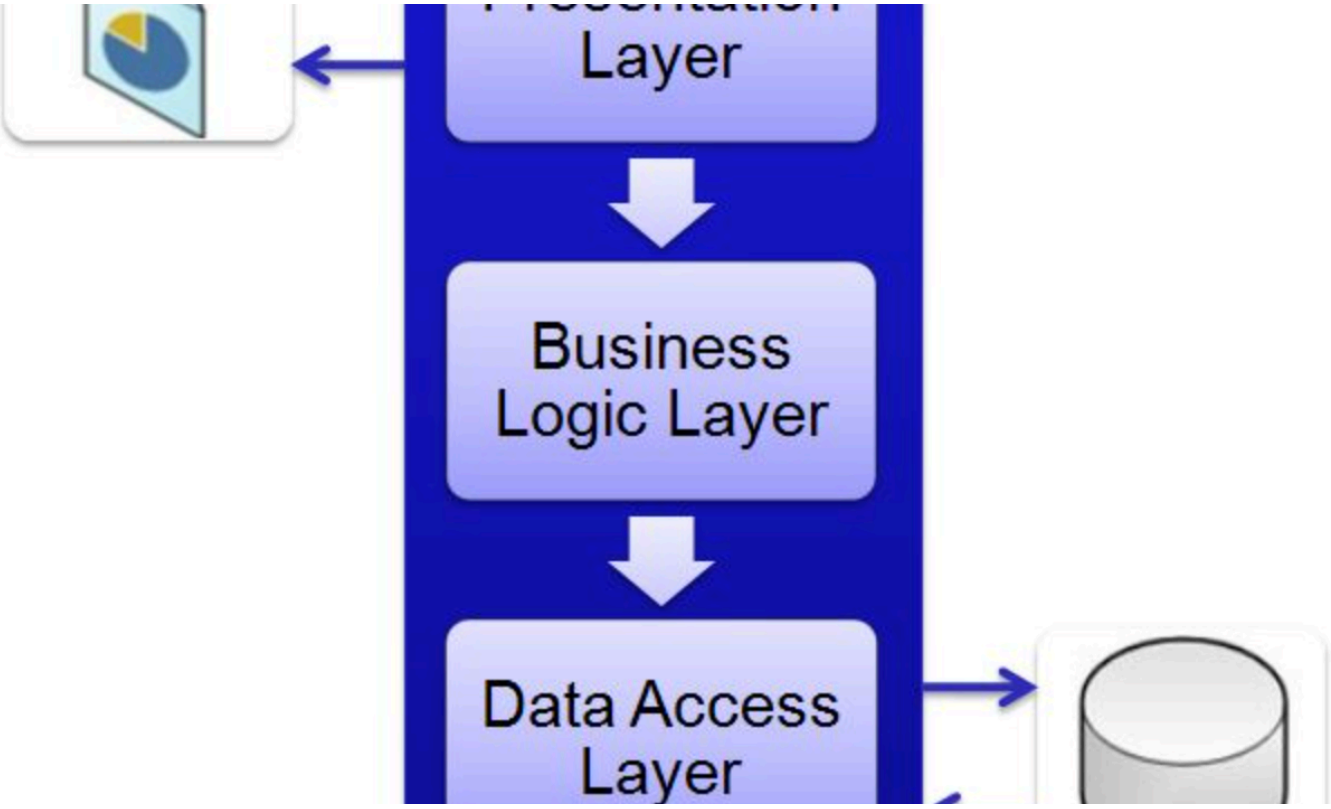
5.Identify and classify 5 applications you use daily as either system software or application software.

- Applications we use daily, classified as either system software or application software: Google Chrome (Application), Google Docs (Application), Windows 11 (System), Zoom (Application), and a PDF reader (Application)

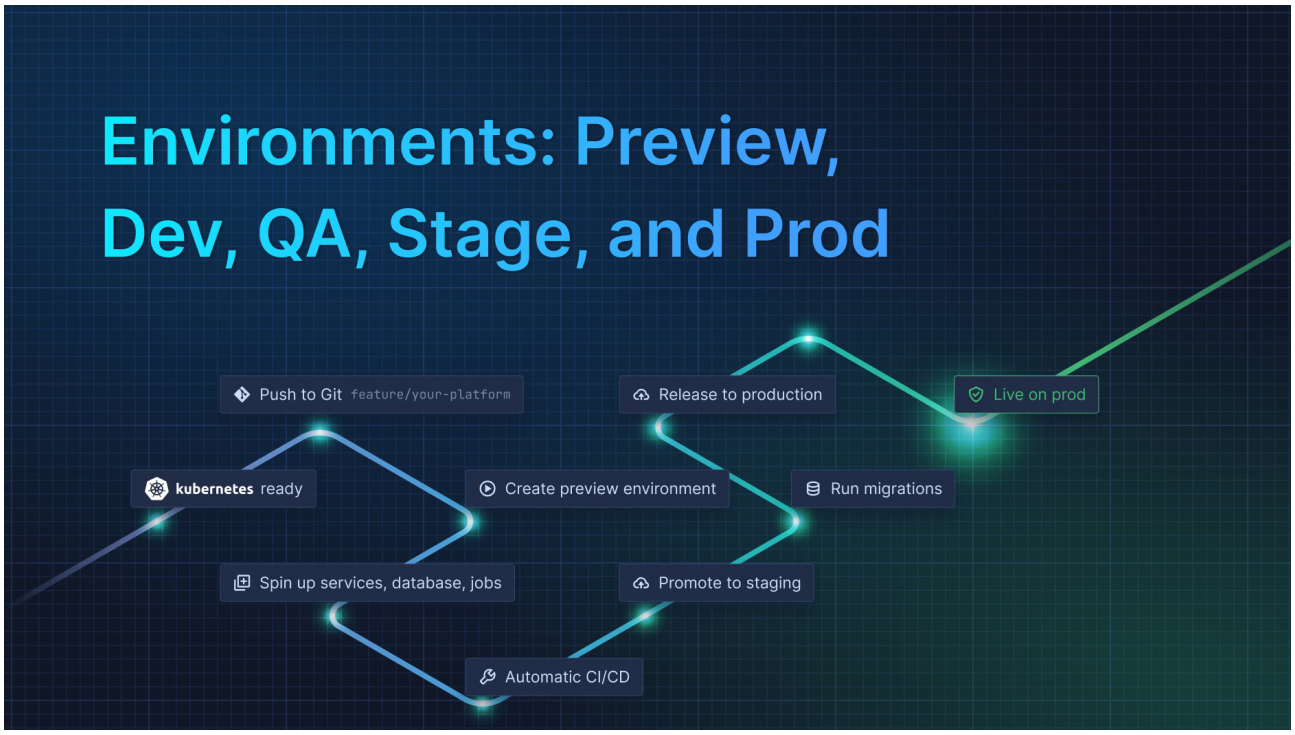
6.Design a basic three-tier software architecture diagram for a web application



7.Create a case study on the functionality of the presentation, business logic, and data access layers of a given software system.

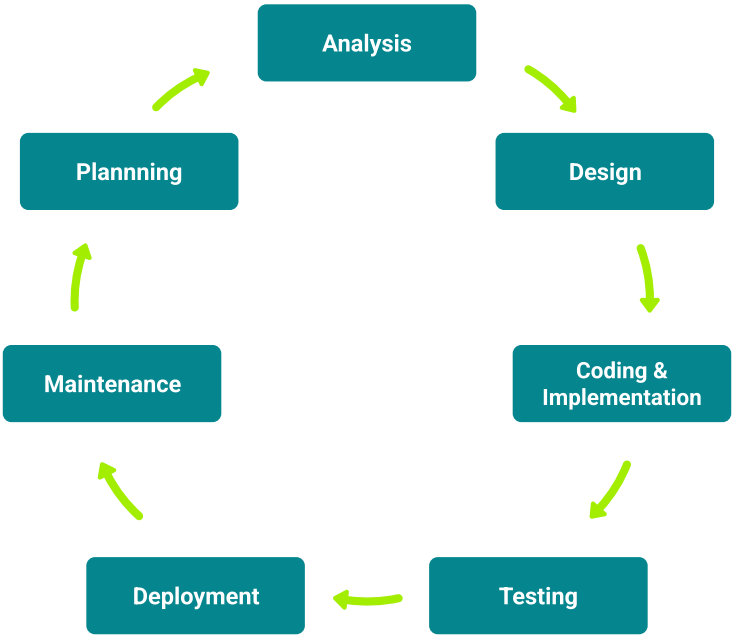


8.Explore different types of software environments (development, testing, production). Set up a basic environment in a virtual machine.



9.Create a flowchart representing the Software Development Life Cycle (SDLC).

Stages of the SDLC



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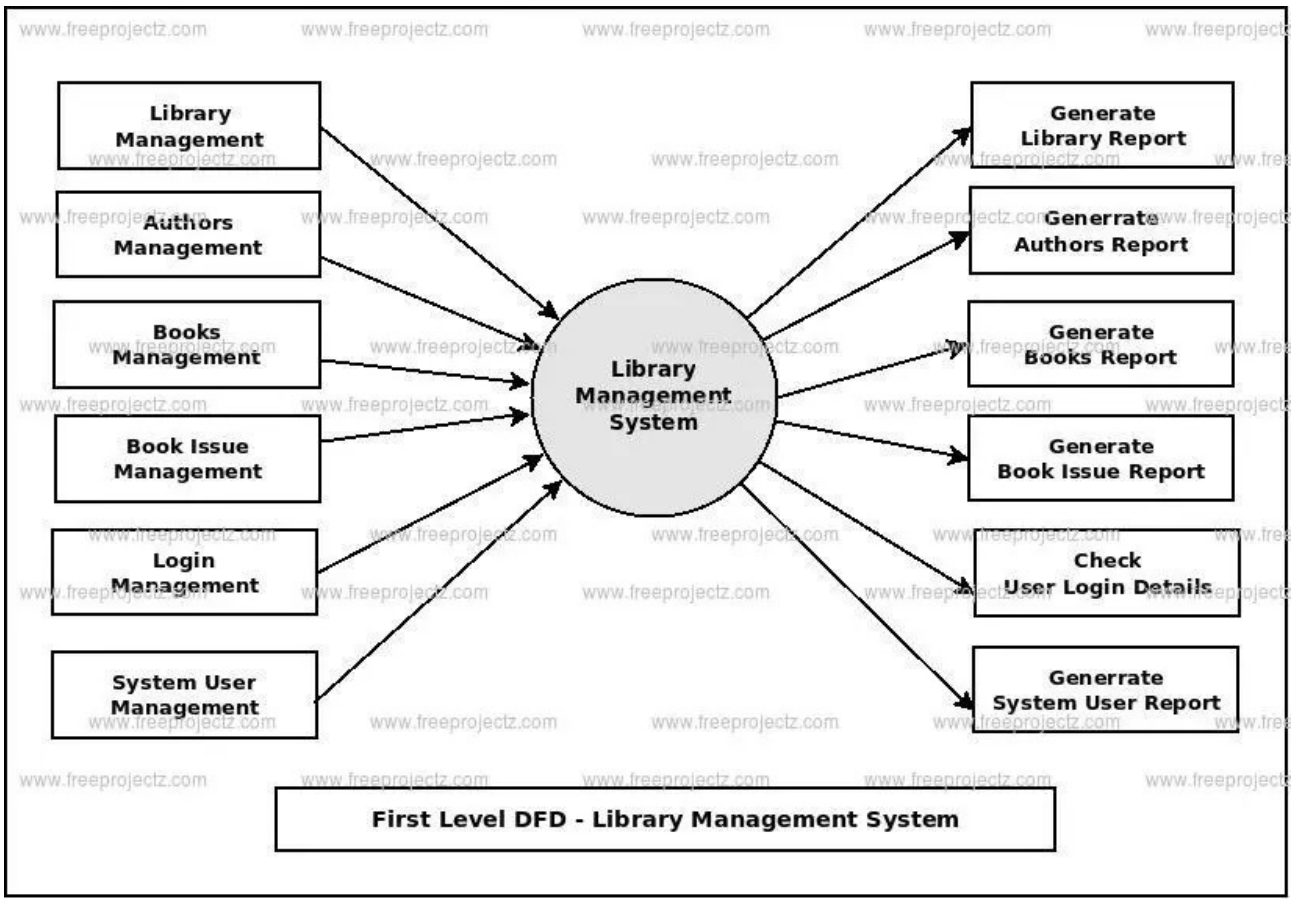
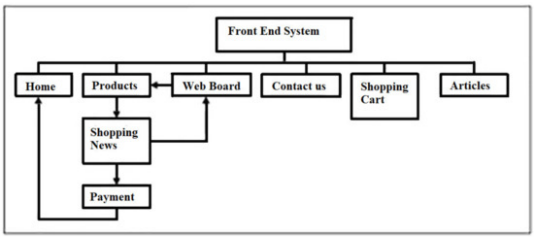
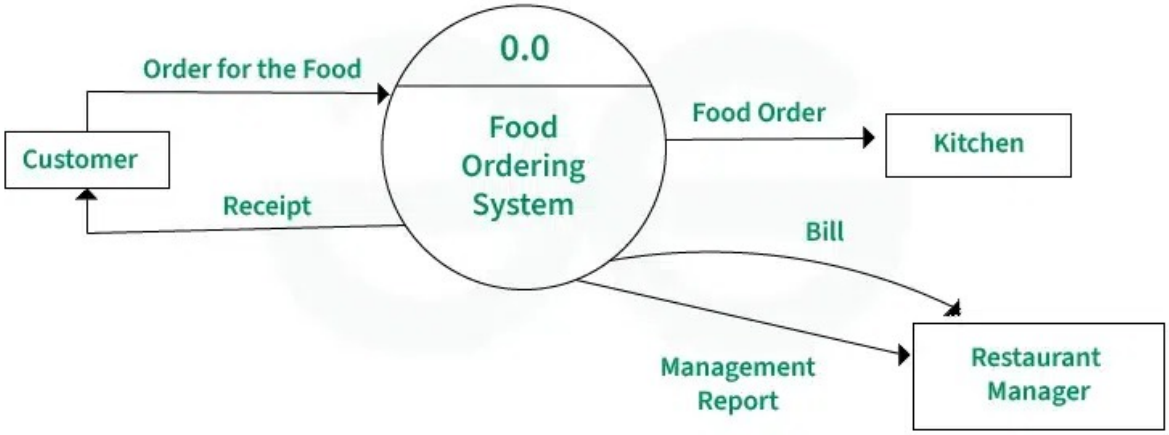
10.Write a requirement specification for a simple library management system



11.Perform a functional analysis for an online shopping system



12.Design a basic system architecture for a food delivery app



DFD level 0

13.Document a real-world case where a software application required critical maintenance